

Comprehensive Business & Vendor Performance Report

Exploratory Data Analysis and Strategic Insights

Generated: July 2026

1. Executive Summary

This report consolidates the findings from the comprehensive Exploratory Data Analysis (EDA) of the `inventory.db` database, evaluating inventory management, purchase orders, and sales performance. A specific focus is placed on the strategic evaluation of **Vendor Performance**, examining profitability, economies of scale (bulk purchasing), and contribution margins.

Key Highlights:

- Analyzed over **15 Million records** across 6 tables, highlighting massive throughput in the supply chain.
- Identified **Top 10 Vendors** that account for over **65%** of the total purchase contribution.
- Discovered significant variance in profit margins among top vendors, ranging from **25% to 37%**.
- Evaluated optimal order sizes, confirming hypotheses around volume-based cost savings.

2. Data Ecosystem & Entity-Relationship (ER) Model

The foundation of this analysis rests on an interconnected SQLite database comprising six core tables. The tables have been cleaned and linked via common identifiers. Below is the structural representation (ER Diagram) showing Primary Keys (**PK**) and Foreign Keys (**FK**).

sales InventoryId (PK) Store Brand (FK) Description Size SalesQuantity SalesDollars SalesPrice VendorNo (FK)	purchases PONumber (PK) VendorNumber (FK) InvoiceDate Brand (FK) Quantity Dollars PurchasePrice Freight	purchase_prices Brand (PK) Description Price Size Volume Classification PurchasePrice VendorNumber (FK)
vendor_invoice InvoiceNumber (PK) VendorNumber (FK)	begin_inventory InventoryId (PK) Store	end_inventory InventoryId (PK) Store

PONumber (FK) InvoiceDate Quantity Dollars Freight	Brand (FK) onHand Price startDate	Brand (FK) onHand Price endDate
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Relationships: 'VendorNumber' links Vendors to Purchases, Sales, and Invoices. 'Brand' acts as a product identifier linking Inventory, Prices, Purchases, and Sales. 'InventoryId' is the unique tracker for stock movement.

3. Exploratory Data Analysis (EDA) Overview

A rigorous examination of the database tables revealed the scale of operations:

- **Sales Records:** 12,825,363 entries, representing the bulk of transactional volume.
- **Purchase Records:** 2,372,474 entries, reflecting supplier acquisitions.
- **Inventory Snapshots:** ~206k beginning records and ~224k ending records.

To facilitate advanced analytics, a consolidated vendor_sales_summary table (10,692 rows) was generated. This aggregated table calculates critical KPIs per vendor and brand, such as **Gross Profit**, **Profit Margin**, **Stock Turnover**, and **Sales-to-Purchase Ratio**.

4. Vendor Performance & Business Outcomes

Top 10 Vendors by Sales & Profitability

The top 10 vendors dominate the procurement budget and sales revenue. The table below illustrates their financial footprint. Notably, while **DIAGEO NORTH AMERICA INC** brings in the highest revenue, vendors like **CONSTELLATION BRANDS INC** and **MARTIGNETTI COMPANIES** yield superior profit margins.

Vendor Name	Total Purchase (\$)	Gross Profit (\$)	Total Sales (\$)	Purchase Contrib. (%)	Profit Margin (%)
DIAGEO NORTH AMERICA INC	50,097,226.16	17,892,873.26	67,990,099.42	16.30	26.32
MARTIGNETTI COMPANIES	25,502,095.83	13,828,263.53	39,330,359.36	8.30	35.16
PERNOD RICARD USA	23,851,164.17	8,212,032.02	32,063,196.19	7.76	25.61
JIM BEAM BRANDS COMPANY	23,494,304.32	7,928,716.14	31,423,020.46	7.64	25.23
BACARDI USA INC	17,432,020.26	7,422,796.88	24,854,817.14	5.67	29.86
CONSTELLATION BRANDS INC	15,273,708.08	8,945,037.57	24,218,745.65	4.97	36.93
E & J GALLO WINERY	12,068,539.22	6,331,360.24	18,399,899.46	3.93	34.41
BROWN-FORMAN CORP	13,238,939.18	5,008,291.47	18,247,230.65	4.31	27.45
ULTRA BEVERAGE COMPANY LLP	11,167,081.61	5,335,462.70	16,502,544.31	3.63	32.33
M S WALKER INC	9,764,312.60	4,942,145.91	14,706,458.51	3.18	33.61

Top Performing Brands

A drill-down into specific brands highlights that consumer preference heavily skews towards staple spirits. Top performers by total sales dollars include:

- **Jack Daniels No 7 Black:** ~\$7.96M
- **Tito's Handmade Vodka:** ~\$7.39M
- **Grey Goose Vodka:** ~\$7.20M
- **Capt Morgan Spiced Rum:** ~\$6.35M

Economies of Scale: Bulk Purchasing vs. Unit Cost

To address the business query: *"Does purchasing in bulk reduce the unit price, and what is the optimal purchase volume for cost savings?"*, the dataset was categorized into three quantiles for **Order Size**: Small, Medium, and Large based on Total Purchase Quantity.

The analysis confirmed that **Larger order sizes correlate with improved unit pricing dynamics and ultimately better profit margins**. High-volume purchases (Large) allow for absorption of fixed costs like freight, significantly lifting the Net Gross Profit. A statistical confidence interval analysis between "Top Vendors" and "Low Vendors" further proves that vendors capable of fulfilling large-scale orders sustainably operate at a higher profit threshold.

5. Strategic Recommendations

- **Vendor Consolidation:** Over 65% of purchasing is concentrated within the top 10 vendors. The procurement team should negotiate aggressive volume-based discount tiers (rebates) with these primary partners (especially DIAGEO and MARTIGNETTI) to lower unit costs further.
- **Margin Optimization:** Re-allocate shelf space and promotional budget towards vendors with higher Profit Margins, such as **CONSTELLATION BRANDS (36.93%)** and **MARTIGNETTI (35.16%)**, rather than purely focusing on top-line revenue drivers like Jim Beam (25.23%).
- **Order Size Strategy:** Shift ordering patterns for fast-moving items (e.g., Jack Daniels, Tito's) from frequent "Medium" orders to consolidated "Large" orders. This strategy exploits economies of scale and dilutes freight and excise tax burdens across a larger quantity of goods.